

Exercise 2.2 — Transition Matrix from Conditional Expectations

Recursive Macroeconomic Theory

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Problem

Consider a two-state Markov chain and a random variable $y_t = \bar{y}^\top x_t$ where

$$\bar{y} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

It is known that $E(y_{t+1}|x_t) = \begin{pmatrix} 1.8 \\ 3.4 \end{pmatrix}$ and that $E(y_{t+1}^2|x_t) = \begin{pmatrix} 5.8 \\ 15.4 \end{pmatrix}$.

We find a transition matrix consistent with these conditional expectations and determine whether the matrix is unique.

Analytical Solution

For a two-state Markov chain, the transition matrix is

$$P = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix}$$

By definition of $P_{ij} = Pr(x_{t+1} = e_j | x_t = e_i)$, we know that $p_{00} + p_{01} = 1$ and that $p_{10} + p_{11} = 1$.

Furthermore, we know that the conditional expectation of a function of next period's state is

$$E(y_{t+1}|x_t) = P\bar{y}.$$

This implies we have to solve the following system in matrix notation:

$$\begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix} \begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 1.8 \\ 3.4 \end{pmatrix},$$

which leads to the following transition matrix:

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

Note that, since the two state values are distinct, the row-sum restriction together with the conditional mean provides two linearly independent equations for the two unknown probabilities in each row. Therefore, the transition matrix is uniquely determined.

Moreover, since $P_{ij} > 0$ for all i, j , Theorem 2.2.1 implies that the Markov chain has a unique stationary distribution and is asymptotically stationary.

Thus, we are only left with checking the second moment condition

$$y^2 = \begin{pmatrix} 1^2 \\ 5^2 \end{pmatrix} = \begin{pmatrix} 1 \\ 25 \end{pmatrix}.$$

Therefore,

$$P \begin{pmatrix} 1 \\ 25 \end{pmatrix} = \begin{pmatrix} 0.8 + 5 \\ 0.4 + 15 \end{pmatrix} = \begin{pmatrix} 5.8 \\ 15.4 \end{pmatrix} = E(y_{t+1}^2 | x_t)$$

Two-State Business-Cycle Model Extension

In Exercise 2.1, the transition matrix was treated as known and used to study the properties of the Markov chain. A natural econometric extension reverses the problem: instead of taking P as given, we seek to recover it from observable implications of the stochastic process. In empirical applications, however, conditional moments are estimated from finite samples and therefore need not satisfy the theoretical restrictions of the model exactly. This motivates a minimum-distance approach.

In particular, think of the economy as being in either a recessionary regime or an expansionary regime, with y_t representing quarterly real GDP growth.

Suppose we approximate GDP growth by two representative values:

$$\bar{y} = \begin{pmatrix} -2 \\ 3 \end{pmatrix},$$

so state 0 means recession and state 1 means expansion. To be clear, these numbers are just illustrative rather than estimates from a particular dataset.

Suppose that, in an empirical application, we obtained the following sample estimates of the conditional moments:

$$E(y_{t+1} | x_t) = \begin{pmatrix} -0.8 \\ 2.4 \end{pmatrix} \quad \text{and} \quad E(y_{t+1}^2 | x_t) = \begin{pmatrix} 5.4 \\ 8.1 \end{pmatrix}.$$

Of course, these sample estimates may not satisfy the theoretical restrictions imposed by the Markov model exactly; this is prevented by sample variation. Thus, instead of demanding a P satisfying every equation exactly, we can look for the transition matrix that comes closest to matching all of the moments simultaneously.

For each current state i , we write

$$q_i = P(x_{t+1} = \text{expansion} | x_t = i).$$

Then,

$$P_{i,\cdot} = (1 - q_i, q_i),$$

and, for any proposed q_i , the model predicts

$$E(y_{t+1} | x_t) = (1 - q_i)(-2) + q_i(3) \quad \text{and} \quad E(y_{t+1}^2 | x_t) = (1 - q_i)4 + q_i9.$$

We choose q_i to minimize

$$(E_{\text{model}}(y_{t+1} | x_t = i) - E_{\text{data}}(y_{t+1} | x_t = i))^2 + (E_{\text{model}}(y_{t+1}^2 | x_t = i) - E_{\text{data}}(y_{t+1}^2 | x_t = i))^2,$$

which is a simple minimum-distance estimator. In the Python companion of this exercise, we estimate the transition matrix to be

$$\hat{P} = \begin{pmatrix} 0.74 & 0.26 \\ 0.15 & 0.85 \end{pmatrix}$$

The estimated matrix implies that, conditional on being in a recession, the economy remains in recession with probability 0.74 and moves into expansion with probability 0.26. Conditional on being in an expansion, the expansion persists with probability 0.85. Unlike the exact solution to Exercise 2.2, $\hat{P}$ does not reproduce every sample moment exactly; rather, it provides the transition matrix that best reconciles the two sets of noisy conditional moments according to the chosen minimum-distance criterion.

Reference

Ljungqvist, L. and Sargent, T. J. (2018). *Recursive Macroeconomic Theory*, 4th edition, MIT Press.